

Notes on General Relativity and Cosmology

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These notes are based on a course taught by Professor Rahulkumar Solanki at Harvey Mudd in Spring 2026. The following additional resources were used to compile these notes.

- *The Classical Theory of Fields* by Landau & Lifshitz. Prof. Solanki's favorite. He followed it quite closely in some sections of the course. Terse introduction to the subject from a legendary series of books.
- *General Relativity* by Wald. Graduate textbook; as Scott Hughes calls it, the "GR überbuch".
- Baumann's notes on cosmology for Part III of Cambridge's mathematical tripos. Recommended to me by Setter.

We use the standard Einstein convention (e.g. $a^i b_i \equiv \sum_{i=1}^3 a^i b_i = a^1 b_1 + a^2 b_2 + a^3 b_3$ for $i \in \{1, 2, 3\}$) and the mostly-plus metric signature throughout these notes.

Contents

1	Principle of equivalence	2
2	The geometry of spacetime	3
2.1	Curvilinear coordinates	3
2.2	Covariant differentiation and Christoffel symbols	4
2.3	Curvature	6
2.4	Geodesics	7
3	Physics	8
3.1	The energy-momentum tensor	8
3.2	Einstein's field equations	10
3.3	Schwarzschild	10
3.4	The Newtonian limit	11
3.5	Newtonian orbits	12
3.6	Measurements	12
3.7	Schwarzschild geodesics	13

1 Principle of equivalence

- In Newtonian mechanics, the inertial mass is defined by $\vec{F} = m_i \vec{a}$ and the gravitational mass is defined by $\vec{F}_g = -m_g \vec{\nabla} \Phi$, where Φ is the gravitational field.
- The *weak equivalence principle* states that $m_i = m_g$, implying that the behavior of freely falling test particles does not depend on their mass. Consequently, the motion of freely falling particles is the same in a uniform gravitational field and a uniformly accelerated frame.
- The *Einstein equivalence principle* generalizes the statement to all local experiments. It tells us that there are no "gravitationally neutral" objects with respect to which we can measure g . Thus we instead define unaccelerated/inertial frames to be freely falling, which motivates us to describe gravity as not a force, but a property of spacetime.
- Since gravitational fields are inhomogeneous, global inertial frames don't exist.
- Consider an inertial frame K and a uniformly rotating frame of reference K' . In a plane perpendicular to the axis of rotation, the ratio of the circumference to the radius of a circle centered at the origin (through which the axis of rotation passes) in the inertial frame K is 2π . However, a meterstick at rest in K' experiences a Lorentz contraction in the azimuthal direction (viewed from inertial frame K) but not in radial direction since it is perpendicular to the relative velocity (once can consider instantaneous inertial frame of reference coinciding with K' at any instance). Therefore, the ratio of circumference to the radius of the same circle is greater than 2π . Thus, the space is non-Euclidean (or curved) in K' ; however, the spacetime is flat (i.e. Minkowski) because there exists a coordinate transformation through which one can return to inertial frame K . Therefore, from the principle of equivalence, gravitational fields have similar property in a small enough region of spacetime. However, "real" gravitational fields cannot be removed by coordinate transformations, so spacetime must be curved in their presence (which can be made locally flat using appropriate coordinate transformations). Thus, the mathematical theory of gravitational fields is the theory of curved spacetime.
- The propagation of light is also affected by gravitational fields since one cannot distinguish uniform acceleration from gravitational field locally using experiments consisting of light signals. In Newtonian theory, propagation of light has nothing to do with gravity. Furthermore, according to special relativity, the mass of a body is a measure of its energy-content; if the energy changes by l , the mass changes in the same sense by l/c^2 . For example, the mass of the hydrogen atom is less than the sum of masses of proton and electron due to negative binding energy. Therefore, according to the equivalence principle, the gravitational mass has to change in exactly the same manner. Essentially, any form of energy (whether mass or electromagnetic field) produces a gravitational field. Finally, the negative binding energy (gravitational or otherwise) makes the theory of general relativity highly nonlinear, since the gravitational field of the end product is not the vector sum of the field produced by constituents.

2 The geometry of spacetime

2.1 Curvilinear coordinates

- Consider the transformation of coordinates $x^\mu \rightarrow x'^\mu$. The basic transformation is that of the differential:

$$dx^i = \frac{\partial x^\mu}{\partial x'^\nu} dx'^\nu$$

- The derivatives of a scalar ϕ transform as

$$\frac{\partial \phi}{\partial x^i} = \frac{\partial \phi}{\partial x'^k} \frac{\partial x'^k}{\partial x^i}$$

- A *contravariant four-vector* is any aggregate of four quantities A^μ transforming like coordinate differentials under transformation of coordinates. The dual four-vectors are the *covariant four-vectors*. Together, these form the set of *rank-1 tensors* in four-dimensional spacetime¹. The transformation laws are

$$A^\mu = \frac{\partial x^\mu}{\partial x'^\nu} A'^\nu \quad (\text{contravariant vector})$$

$$A_\mu = \frac{\partial x'^\nu}{\partial x^\mu} A'_\nu \quad (\text{covariant vector})$$

- Rank-2 tensors* are aggregates of 16 quantities transforming like

$$A^{\mu\nu} = \frac{\partial x^\mu}{\partial x'^\rho} \frac{\partial x^\nu}{\partial x'^\sigma} A'^{\rho\sigma} \quad (\text{contravariant})$$

$$A_{\mu\nu} = \frac{\partial x'^\rho}{\partial x^\mu} \frac{\partial x'^\sigma}{\partial x^\nu} A'_{\rho\sigma} \quad (\text{covariant})$$

$$A^\mu_\nu = \frac{\partial x^\mu}{\partial x'^\rho} \frac{\partial x'^\sigma}{\partial x^\nu} A'^\rho_\sigma \quad (\text{mixed})$$

- More generally, a tensor of type (r, s) , or rank $r + s$, is an object with r contravariant indices and s covariant indices whose components transform with one Jacobian factor for each index:

$$A^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} = \frac{\partial x^{\mu_1}}{\partial x'^{\rho_1}} \dots \frac{\partial x^{\mu_r}}{\partial x'^{\rho_r}} \frac{\partial x'^{\sigma_1}}{\partial x^{\nu_1}} \dots \frac{\partial x'^{\sigma_s}}{\partial x^{\nu_s}} A'^{\rho_1 \dots \rho_r}_{\sigma_1 \dots \sigma_s}$$

Thus each upper index transforms like a contravariant vector index, and each lower index transforms like a covariant vector index.

- The *Kronecker delta* δ^μ_ν is a (rank-2) tensor which can "replace indices" like $A^\nu \delta^\mu_\nu = A^\mu$. Note:

$$\frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x^\mu}{\partial x'^\beta} = \delta^\alpha_\beta$$

$$\frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x'^\mu}{\partial x^\sigma} = \delta^\rho_\sigma$$

¹From now on, when we speak of tensors, we assume that we are talking about tensors in four-dimensional spacetime, using the physicist's conventions.

- A spacetime is called *flat* if there exists a coordinate system in which the metric takes the form² $g_{\mu\nu} = g^{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ everywhere. Such coordinates are called *galilean coordinates* or *global inertial coordinates*.
- Spacetimes coordinates x'^μ are said to be *curvilinear* if they are related to Galilean coordinates x^μ by a smooth, locally invertible transformation; generally, this transformation is nonlinear, so the coordinate lines need not be straight.
- An infinitesimal *spacetime interval* between two nearby events in curvilinear coordinates is

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

where $g_{\mu\nu}$ are functions of the spacetime coordinates. $g_{\mu\nu}$ forms a symmetric covariant tensor called the *metric tensor*. The metric tensor raises/lowers indices: $A^\mu = g^{\mu\nu} A_\nu$, $A_\mu = g_{\mu\nu} A^\nu$. This also applies for tensors of higher rank; $A^\mu_\nu = g^{\mu\rho} A_{\rho\nu}$, $A^{\mu\nu} = g^{\mu\rho} g^{\nu\sigma} A_{\rho\sigma}$. In particular, $g_{\alpha\mu} g^{\beta\mu} = \delta^\alpha_\beta$.

- Let x'^μ be galilean coordinates and x^μ arbitrary curvilinear coordinates. Consider the Jacobian of the transformation from the galilean to curvilinear coordinates $J = \frac{\partial(x^0, x^1, x^2, x^3)}{\partial(x'^0, x'^1, x'^2, x'^3)}$. Then the determinant g of the metric tensor $g_{\mu\nu}$ in the system x^μ is given by $J = \frac{1}{\sqrt{-g}}$. So for example, in a galilean coordinate system the integral of a scalar with respect to the four-volume element $d\Omega' = dx'^0 dx'^1 dx'^2 dx'^3$ is also a scalar, so on transforming to curvilinear coordinates, the element of integration goes over into $d\Omega' \rightarrow \sqrt{-g} d\Omega$. Thus, in curvilinear coordinates, when integrating over a four-volume, the quantity $\sqrt{-g} d\Omega$ behaves like an invariant.
- From now on, we denote by τ the *proper time* (i.e. the time experienced in an object's rest frame). For two infinitesimally separated events happening at the same point in space, and denoting by $d\tau$ the (proper) time interval between the two events, we have $ds^2 = -c^2 d\tau^2$, from which $d\tau = \frac{1}{c} \sqrt{-g_{00}} dx^0$. So, the (proper) time between any two events occurring at the same point in space is

$$\tau = \frac{1}{c} \int \sqrt{-g_{00}} dx^0$$

2.2 Covariant differentiation and Christoffel symbols

In galilean coordinates, the differential dA^μ of a vector A^μ forms a vector, and the derivatives $\frac{\partial A^\mu}{\partial x^\nu}$ form a tensor. In general curvilinear coordinates, neither quantity is tensorial, due to the fact that dA^μ is the difference of vectors located at different, infinitesimally separated points of space. In order to obtain a differential of a vector that behaves like a vector. It is necessary that the two vectors to be subtracted from each other be located at the same point in space; that is, we translate one of the vectors to the point where the second is, after which we determine the difference of two vectors which we now refer to one and the same point in space. We also need the operation of translation to be defined so that in galilean coordinates the difference coincides with dA^μ . Under parallel translation of a vector, its components in galilean don't change, but in curvilinear coordinates they generally do (the canonical example is a vector on the equator of a sphere, parallel transported to one of the poles, then back down to the equator in a direction perpendicular to the first displacement, and finally back to the original point on the equator; the

²This is the so called "east coast convention" used in general relativity. In quantum field theory, and in many treatments of general relativity, one uses the metric tensor $g_{\mu\nu} = g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

final vector comes back rotated by 90 degrees). This ends up being precisely how we do calculus in curvilinear coordinates.

- The change in a vector under a *parallel displacement* is given by

$$DA^\mu = \left(\frac{\partial A^\mu}{\partial x^\rho} + \Gamma_{\nu\rho}^\mu A^\nu \right) dx^\rho$$

$$DA_\mu = \left(\frac{\partial A_\mu}{\partial x^\rho} - \Gamma_{\mu\rho}^\nu A_\nu \right) dx^\rho$$

where the quantities $\Gamma_{\nu\rho}^\mu$ are (local) coefficients called *Christoffel symbols*. The expressions in the parentheses, called the *covariant derivatives* of vectors A^μ , A_μ , denoted $\nabla_\rho A^\mu$ and $\nabla_\rho A_\mu$ respectively, give the desired generalization of the concept of a derivative to curvilinear coordinates.

- One can similarly determine the covariant derivative of a tensor of arbitrary rank; to obtain the covariant derivative of a tensor $A_{\dots}$ with respect to x^ρ , we add to the ordinary derivative $\partial A_{\dots}/\partial x^\rho$ for each covariant index ($A_{\dots\mu}$) a term $-\Gamma_{\mu\rho}^\nu A_{\dots\nu}$, and for each contravariant index ($A^{\dots\mu}$) a term $+\Gamma_{\nu\rho}^\mu A^{\dots\nu}$.
- In a Galilean coordinate system, the Christoffel symbols vanish, and covariant differentiation reduces to ordinary differentiation.
- In terms of the metric tensor,

$$\Gamma_{\nu\rho}^\mu = \frac{g^{\mu\sigma}}{2} (\partial_\nu g_{\rho\sigma} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho}) = \Gamma_{\rho\nu}^\mu$$

- The Christoffel symbols are not tensors: they behave like a tensor only under linear transformations. Generally, they transform as

$$\Gamma_{\mu\lambda}^{\nu'} = \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\lambda} \Gamma_{\rho\sigma}^\alpha - \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\lambda} \frac{\partial^2 x^{\nu'}}{\partial x^\alpha \partial x^\beta}$$

- The covariant derivative as defined above indeed follows the conditions one requires of a derivative operator: linearity, the Leibnitz (product) rule, commutativity with contraction, consistency with the notion of tangent vectors as directional derivatives on scalar fields, and finally, the "torsion-free" condition $\nabla_\mu \nabla_\nu f = \nabla_\nu \nabla_\mu f$ for scalar f . While this last condition is sometimes dropped, in general relativity the derivative operator is assumed to be torsion-free.
- The metric tensor acts as a constant under covariant differentiation, that is, $\nabla_\rho g_{\mu\nu} = 0$; in fact, for any given metric, the derivative operator ∇_ρ satisfying this condition exists and is unique.
- If in a covariant derivative we raise the index signifying the differentiation, we obtain the so-called *contravariant derivative*; thus, $\nabla^\nu A_\mu = g^{\nu\rho} \nabla_\rho A_\mu$, $\nabla^\nu A^\mu = g^{\nu\rho} \nabla_\rho A^\mu$.
- It is always possible to choose a coordinate system in which all the $\Gamma_{\nu\rho}^\mu$ become zero at a previously assigned point. Such a system is said to be *locally inertial*. In fact, let a given point be chosen as the origin of coordinates, and let the values of $\Gamma_{\nu\rho}^\mu$ at it be initially (in the coordinates x^μ) equal to $(\Gamma_{\nu\rho}^\mu)_0$. In the neighborhood of this point, make the transformation $x'^\mu = x^\mu + \frac{1}{2} (\Gamma_{\nu\rho}^\mu)_0 x^\nu x^\rho$. Then all the $\Gamma_{\nu\rho}^\mu$ become zero. Such a transformation has $\left(\frac{\partial x'^\mu}{\partial x^\nu} \right)_0 = \delta_\nu^\mu$, so it does not change the value of any tensor at the given point, so we can make the Christoffel symbols vanish at the same time as we bring the $g_{\mu\nu}$ to Cartesian form.

- A vector A^μ given at each point on some curve C with a tangent B^μ is said to be *parallelly transported* as one moves along the curve if $B^\mu \nabla_\mu A^\nu = 0$ along the curve. More generally, one can define the parallel transport of a tensor of arbitrary rank by $B^\mu \nabla_\mu A_{\sigma_1 \dots \sigma_s}^{\rho_1 \dots \rho_r}$. Choosing a coordinate basis and parametrizing C by t , $\frac{dA^\nu}{dt} + B^\mu \Gamma_{\mu\lambda}^\nu A^\lambda = 0$.

2.3 Curvature

Euclid's fifth postulate states that if two lines are drawn which intersect a third in such a way that the sum of the inner angles on one side is less than two right angles, then the two lines inevitably must intersect each other on that side if extended far enough. Two-dimensional space of constant negative curvature satisfies all but this postulate of Euclid. There are two approaches to studying the curvature of an object: the *extrinsic approach*, which is to analyze the object by embedding it in a higher-dimensional space, or to analyze it *intrinsically* by measuring entirely from within, without referencing any higher-dimensional surrounding space. General relativity studies curved spacetime intrinsically.

- Let $\delta A_\mu = \Gamma_{\mu\nu}^\gamma A_\gamma dx^\nu$ be the change in DA_μ under the parallel transport from $x^\mu \rightarrow x^\mu + dx^\mu$, i.e. $DA_\mu = dA_\mu + \delta A_\mu$. This δA_μ is path-dependent due to the dx^ν term, and therefore so is the covariant difference. The change in an arbitrary vector field after parallel displacement around any infinitesimal closed contour gives us necessary and sufficient conditions for whether or not spacetime is curved. Write this change $\Delta A_\mu = \oint \delta A_\mu = \oint \Gamma_{\mu\nu}^\lambda A_\lambda dx^\nu$. Then, with the first-order accuracy sufficient for the transformation, $\Delta A_\mu = \frac{1}{2} R_{\mu\nu\sigma}^\lambda A_\lambda \Delta f^{\nu\sigma}$, where $\Delta f^{\nu\sigma}$ is the infinitesimal value of the area enclosed by the contour, and $R_{\mu\nu\sigma}^\lambda$ is the curvature tensor defined below.
- Define the *curvature tensor* (or *Riemann tensor*)

$$R_{\mu\nu\alpha\beta} u^\nu = (\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) A_\mu$$

Equivalently,

$$\begin{aligned} R_{\nu\rho\sigma}^\mu &= \partial_\rho \Gamma_{\nu\sigma}^\mu - \partial_\sigma \Gamma_{\nu\rho}^\mu + \Gamma_{\alpha\rho}^\mu \Gamma_{\nu\sigma}^\alpha - \Gamma_{\alpha\sigma}^\mu \Gamma_{\nu\rho}^\alpha \\ R_{\mu\nu\rho\sigma} &= \frac{1}{2} (\partial_\nu \partial_\rho g_{\mu\sigma} + \partial_\mu \partial_\sigma g_{\nu\rho} - \partial_\nu \partial_\sigma g_{\mu\rho} - \partial_\mu \partial_\rho g_{\nu\sigma}) + g_{\alpha\beta} (\Gamma_{\nu\rho}^\alpha \Gamma_{\mu\sigma}^\beta - \Gamma_{\nu\sigma}^\alpha \Gamma_{\mu\rho}^\beta) \end{aligned}$$

This tensor vanishes if and only if the curvature is zero. By way of coordinate transformations, we can make $\Gamma_{\beta\sigma}^\alpha = 0$ at a single point, but $\partial_\beta \Gamma_{\beta\gamma}^\alpha \neq 0$ if the spacetime is curved.

- Properties of the Riemann tensor include:

$$\begin{aligned} R_{\alpha\beta\gamma\delta} &= -R_{\beta\alpha\gamma\delta} = R_{\beta\alpha\delta\gamma} = R_{\gamma\delta\alpha\beta} g^{\mu\nu} R_{\mu\nu\rho\sigma} = R_{\mu\rho\sigma}^\mu = 0 \\ R_{\alpha\beta\gamma\delta} + R_{\beta\gamma\alpha\delta} + R_{\gamma\alpha\beta\delta} &= 0 \\ \nabla_\alpha R_{\beta\gamma\delta\sigma} + \nabla_\beta R_{\gamma\alpha\delta\sigma} + \nabla_\gamma R_{\alpha\beta\delta\sigma} &= 0 \end{aligned}$$

- The trace of the curvature (or Riemann) tensor is captured by the *Ricci tensor* $R_{\nu\sigma} = g^{\mu\rho} R_{\mu\nu\rho\sigma} = R_{\nu\mu\sigma}^\mu$ and tracefree part is captured by the *Weyl tensor* $C_{\alpha\beta\gamma\delta} = R_{\alpha\beta\gamma\delta} - \frac{1}{2} R_{\alpha\gamma} g_{\beta\delta} + \frac{1}{2} R_{\alpha\delta} g_{\beta\gamma} + \frac{1}{2} R_{\beta\gamma} g_{\alpha\delta} - \frac{1}{2} R_{\beta\delta} g_{\alpha\gamma} + \frac{R}{6} (g_{\alpha\gamma} g_{\beta\delta} - g_{\alpha\delta} g_{\beta\gamma})$. In some notations, one writes $C_{\rho\sigma}^{\mu\nu} = R^{\mu\nu}{}_{\rho\sigma} - 2\delta_{[\rho}^{[\mu} R_{\sigma]}^{\nu]} + \frac{R}{3} \delta_{[\rho}^{[\mu} \delta_{\sigma]}^{\nu]}$, where the square bracket around indices represents antisymmetrization $A_{[\alpha\beta]} = \frac{1}{2} (A_{\alpha\beta} - A_{\beta\alpha})$.
- The *Ricci scalar* is $g^{\nu\sigma} R_{\nu\sigma} = R$.

2.4 Geodesics

- The equations of motion for a free motion of a particle in special relativity are $\frac{du^\mu}{d\tau} = 0$ or $du^\mu = 0$, where $u^\mu = \frac{dx^\mu}{d\tau}$ is the *four-velocity*. It follows by definition that $u^\mu u_\mu = -1$.
- We postulate that the motion of freely falling massive particles (i.e. particles traveling with speed less than c under the influence of only gravity) is given by the path of maximum proper time, called a *timelike geodesic*. Physically, this postulate makes sense because such paths are locally straight through spacetime, just like segments of minimum length are straight lines in Euclidean space. A physical approach to finding this motion would be to set $a^\mu = \frac{Du^\mu}{D\tau} = 0$. Equivalently, such particles satisfy the equations of motion

$$\frac{du^\mu}{d\tau} + \Gamma_{\nu\rho}^\mu u^\nu u^\rho = \frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0$$

We call this the *timelike geodesic equation*.

- Alternatively, we can define a timelike geodesic to be a timelike curve $x^\mu = x^\mu(\sigma)$ joining two events A and B , where σ is some parameter, satisfying

$$\delta S = \delta \left(\int_A^B \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} d\sigma \right) = 0$$

among all the timelike curves in the vicinity joining A and B . Recall that the tangent to such a timelike curve is $\frac{dx^\mu}{d\sigma}$. Curves satisfying the above then follow the Euler-Lagrange equations $\frac{\partial \mathcal{L}}{\partial x^\alpha} - \frac{d}{d\sigma} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha} \right) = 0$ where $L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}}$; then the first definition is recovered.

- A spacelike path cannot be travelled by massive or massless particles since the events lying on these paths are non-causal. However, spacelike geodesics can be useful in defining spatial coordinates, for example, in the case of Fermi normal coordinates. If σ is a parameter for a spacelike curve $x^\mu = x^\mu(\sigma)$, then $g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} > 0$ and one can proceed in a similar manner as the previous parts to find a geodesic equation with proper length s (instead of τ):

$$\frac{d^2 x^\mu}{ds^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{ds} \frac{dx^\rho}{ds} = 0$$

- For a null curve, $ds^2 = 0$, so it does not make sense to apply the variational principle since all null paths will give total spacetime interval between fixed events A, B to equal zero. If $x^\mu = x^\mu(\lambda)$ is a null curve, then one generally chooses λ such that the wave four-vector is $k^\mu = \frac{dx^\mu}{d\lambda}$. Since we consider geodesics as generalizations of straight lines, $Dk^\mu = 0$ is the geodesic equation. We have thus obtained the *lightlike geodesic equation*

$$\frac{dk^\mu}{d\lambda} + \Gamma_{\nu\sigma}^\mu k^\nu k^\sigma = 0 = \frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\sigma}{d\lambda}$$

- It is very important to understand the difference between a null path and a null geodesic; for a null path, $k^\mu k_\mu = 0$, and null geodesics satisfy the geodesic equation in addition to the null path condition.

3 Physics

3.1 The energy-momentum tensor

- The *(energy-)momentum four-vector* p^μ of a particle of mass m is defined by $p^\mu = mu^\mu$. The *energy* of a particle as measured by an observer present at the site of the particle whose 4-velocity is v^μ is defined by $E = -p_\mu v^\mu$. Thus, in special relativity, energy is recognized to be the time component of p^μ . For a particle at rest with respect to the observer (i.e. $v^\mu = u^\mu$), this reduces to the familiar $E = mc^2$. From now on, we use natural units, setting $c = 1$.
- We find that $p^\mu = mu^\mu$, $p^\mu p_\mu = -m^2$.
- Continuous matter distributions in special relativity are described by a symmetric (*stress-energy-momentum* tensor).
- Consider the Lorentz transformation

$$\begin{aligned} x'^0 &= \frac{x^0 - ux^1}{\sqrt{1 - u^2}} & x'^1 &= \frac{x^1 - ux^0}{\sqrt{1 - u^2}} \\ x'^2 &= x^2 & x'^3 &= x^3 \end{aligned}$$

moving from a frame S to a frame S' moving at a velocity u in the $+x$ direction relative to S . The line element $dl^2 = -dt^2 + dx^2 + dy^2 + dz^2$ is invariant under such transformations.

- Under such transformations, a four-vector A^μ transforms as

$$\begin{aligned} A'^0 &= \frac{A^0 - uA^1}{\sqrt{1 - u^2}} & A'^1 &= \frac{A^1 - uA^0}{\sqrt{1 - u^2}} \\ A'^2 &= A^2 & A'^3 &= A^3 \end{aligned}$$

- Unlike mass, density is not an invariant, so it transforms under Lorentz transformations. For the treatment of charge density (which is equivalent to that of number density), consider the charge enclosed by some volume V in S . When we Lorentz-transform, the charge in the corresponding volume V' in S' is equal to the charge enclosed by V . Thus $\rho dV = \rho' dV' = \rho_0 dV_0$, where the last quantity refers to the proper frame, i.e. where the charges are at rest. Let v, v' denote the velocities of S, S' w.r.t. the proper frame, respectively. Then:

$$\begin{aligned} dV &= dV_0 \sqrt{1 - v^2} & \rho &= \frac{\rho_0}{\sqrt{1 - v^2}} \\ dV' &= dV_0 \sqrt{1 - v'^2} & \rho' &= \frac{\rho_0}{\sqrt{1 - v'^2}} \end{aligned}$$

yielding $\rho' = \rho \sqrt{\frac{1-v^2}{1-v'^2}}$. Recalling velocity transformations $v' = \frac{v-u}{1-uv}$,

$$\rho' = \frac{\rho(1-uv)}{\sqrt{1-u^2}} = \frac{\rho - u(\rho v)}{\sqrt{1-u^2}}$$

- $\vec{J} = \rho \vec{v}$ is the *charge flux* or *current density*, i.e. the charges per unit area per unit time.
- Now consider a case in which the charges are in S , moving in a direction normal to some fixed cross sectional area A at a velocity v . In time dt , the amount of charges passing through A is given by $\rho A v dt$, so the charge flux is ρv . Now, $\rho' v' = \frac{\rho(1-uv)}{\sqrt{1-u^2}} \left(\frac{v-u}{1-uv} \right) = \frac{\rho(v-u)}{\sqrt{1-u^2}}$.

- We construct the *four-current density*

$$J^\mu = (\rho, \vec{J}) = (\rho, \rho \vec{v}) = \rho_0 \left(\frac{1}{\sqrt{1-v^2}}, \frac{\vec{v}}{\sqrt{1-v^2}} \right) = \rho_0 u^\mu$$

- A *global conservation* law for some quantity (e.g. for charge or energy) tells us that the integral of that quantity's density over all volume is constant. Then *local conservation* needs to hold for causality; this yields a *continuity* condition that is, for some arbitrary volume V enclosed by a surface S , local conservation requires that the rate of change of total charges inside V equal the number of charges leaving or entering surface S per unit time. Mathematically,

$$\begin{aligned} -\frac{\partial}{\partial t} \iiint_V \rho dV &= \oint_S \vec{J} \cdot d\vec{S} \\ \Rightarrow \iiint_V -\frac{\partial \rho}{\partial t} dV &= \iiint_V (\nabla \cdot \vec{J}) dV \\ \Rightarrow \frac{\partial S}{\partial t} + \nabla \cdot \vec{J} &= 0 \end{aligned}$$

In terms of four-vectors, we can write $\frac{\partial J^\mu}{\partial x^\mu} = 0$. The covariant version is simply $\nabla_\mu J^\mu = 0$.

- For energy density, we have the same setup. Because energy is a component of the four-momentum, which is a four-vector, it transforms under a Lorentz transform accordingly:

$$E' = \frac{E - up_x}{\sqrt{1-u^2}}, \quad p'_x = \frac{p_x - uE}{\sqrt{1-u^2}}$$

- Energy flux in the $+x$ direction is the amount of energy passing through unit area in yz plane in unit time. Thus energy flux equals energy density $\frac{\rho_0}{\sqrt{1-v^2}}$ times velocity; this is just then momentum density. This gives us the *energy-momentum tensor for pressureless dust*:

$$T^{\mu\nu} = \rho u^\mu u^\nu$$

- For a particle, a rank 1 four-momentum tells us how energy and momentum transform from one frame to another; for a continuous distribution of matter, a rank 2 energy-momentum tensor tells us how energy and momentum densities transform from one frame to another.
- T^{00} is the energy density, T^{i0} is the (3-)momentum density, T^{0i} is the energy flux, and T^{ik} is the stress tensor or momentum flux. $T^{\mu\nu} = T^{\nu\mu}$ due to energy flux equaling momentum density and the stress tensor being symmetric (due to conservation of angular momentum).
- In terms of $T^{\mu\nu}$, conservation of energy is $\nabla_\mu T^{0\mu} = 0$, and conservation of momentum (in the i th direction) is $\nabla_\mu T^{i\mu} = 0$. Thus, in general, $\nabla_\mu T^{\mu\nu} = 0$.
- Instead of pressureless dust, consider a perfect fluid with isotropic pressure. In the rest/proper/comoving frame of the fluid, there is only isotropic pressure P and no dissipation or energy flux. Therefore, for a perfect fluid in the rest frame,

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

In general,

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + Pg^{\mu\nu}$$

The electromagnetic field behaves somewhat like a pressurized fluid; it turns out that for an electromagnetic field $T^{\mu\nu} = F^{\mu\alpha}F^\nu_\alpha - \frac{g^{\mu\nu}}{4}F_{\alpha\beta}F^{\alpha\beta}$ where $F^{\mu\nu}$ is the electromagnetic tensor.

- The most general form of the energy-momentum tensor is

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + q^\mu u^\nu + q^\nu u^\mu + Pg^{\mu\nu} + \pi^{\mu\nu}$$

where q^μ is energy flux and $\pi^{\mu\nu}$ is (traceless) anisotropic pressure.

3.2 Einstein's field equations

- Matter tells spacetime how to curve and spacetime tells matter how to move. The curvature of spacetime should then correspond to the energy-momentum of matter/fluid. But the latter is (4-)divergence-free, i.e. $\nabla_\mu T^{\mu\nu} = 0$, whereas the rank-2 tensor we use to describe curvature—the Ricci tensor—is not divergence-free. Using the Bianchi identity, one defines a divergence-free tensor, called the *Einstein tensor*:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{R}{2}g_{\mu\nu}$$

- *Einstein's field equations* are

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

- The formal classical field theory approach to derive this is to use the *Einstein-Hilbert* action

$$S = S_{\text{gravity}} + S_{\text{matter}} = \int R\sqrt{-g}d\Omega + \int T_{\mu\nu}\delta g^{\mu\nu}\sqrt{-g}d\Omega$$

where $g = \det(g_{\mu\nu})$ and $d\Omega$ is a four-volume element. This can be compared to classical electrodynamics, where

$$S = S_{\text{field}} + S_{\text{interactions}} + S_{\text{matter}} = \int F_{\mu\nu}F^{\mu\nu}\sqrt{-g}d\Omega + \int A_\mu J^\mu\sqrt{-g}d\Omega + \int \mu\sqrt{-g}d\Omega$$

where μ is mass density. The equation $\delta S_{\text{interactions}} + \delta S_{\text{matter}} = 0$ gives the Lorentz force $F^{\alpha\beta}J_\beta = \mu a^\alpha$, and $\delta S_{\text{field}} + \delta S_{\text{interactions}} = 0$ gives the field equation $\nabla_\mu F^{\mu\nu} = -J^\nu$.

- Rank 2 tensors have 16 components. Symmetric rank 2 tensors thus have 10 independent components. Conservation of energy $\nabla_\mu T^{\mu\nu} = 0$ means there are 4 constraints or 4 degrees of freedom that can be gauge fixed. Thus, there are truly 6 independent field equations.

3.3 Schwarzschild

- The first *exact* solution of Einstein's field equation was found by Schwarzschild. He gave two solutions, interior and exterior, where the former refers to the gravitational field inside a spherically symmetric constant density star, whereas the latter refers to the gravitational field outside any spherically symmetric star, that is, a vacuum solution.
- In vacuum, $T_{\mu\nu} = 0 = R_{\mu\nu}$, thus $R = 0$.

- A solution is called *asymptotically flat* if, infinitely far away from the spherical body producing the gravitational field, the spacetime becomes flat/Minkowski.
- A solution is called *static* if spacetime does not change over time and is also irrotational. This is a spatial case of *stationary spacetime*, in which the geometry does not change in time but can rotate.
- *Birkhoff's theorem* states that any spherically symmetric solution of the vacuum field equations (e.g. the Schwarzschild exterior solution, which is unique) is static and asymptotically flat.
- The solution being static means that spherically symmetric gravitational waves don't exist, even though matter in spherical stars is allowed to move/oscillate radially.
- We consider the static solution. A static solution has a line element of the form

$$ds^2 = g_{tt}dt^2 + g_{ik}dx^i dx^k$$

where $\partial_t g_{\mu\nu} = 0$.

- The most general spherically symmetric static line element is

$$ds^2 = -e^{2v(r)}dt^2 + e^{2\lambda(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

The Einstein tensor is then

$$G_t^t = -\frac{e^{2\lambda}}{r^2}(2r\lambda' + e^{2\lambda} - 1) \quad G_r^r = -\frac{e^{2\lambda}}{r^2}(e^{2\lambda} - 1 - 2rv')$$

$G_\theta^\theta = G_\phi^\phi \neq 0$ but we don't need it to find $\lambda(r), v(r)$. Now, $G_t^t = 0 \implies \frac{d}{dr}(re^{-2\lambda}) = 1 \implies e^{-2\lambda} = (1 - \frac{r_s}{r})$, where r is an integration constant; furthermore, $G_r^r = 0$ and $e^{2v} = (1 - \frac{r_s}{r})$. We take $r_s = 2M$ for reasons which will be clear in the next section. Basically, in the weak field limit, M is mass or newtonian mass.

- Thus the exterior Schwarzschild solution reduces to

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \frac{dr^2}{\left(1 - \frac{2M}{r}\right)} + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

If one inserts G and c back then $|g_{tt}| = 1 - \frac{2GM}{rc^2}$.

3.4 The Newtonian limit

- Consider a spherically symmetric gravitational field produced by a star in Newtonian gravity. The equation of motion of a test particle in a radial ingoing free fall is given by $\vec{a} = -\nabla\Phi \implies \frac{d^2r}{dt^2} = \frac{M}{r^2}$ where we take $G = 1$.
- The same notion in general relativity is the geodesic equation with $\frac{d\theta}{d\tau} = \frac{d\phi}{d\tau} = 0$. We consider

equatorial plane without loss of generality i.e. $\theta = \pi/2$. One computes the Christoffel symbols

$$\begin{aligned}\Gamma_{tt}^r &= \frac{M}{r^2} \left(1 - \frac{2M}{r}\right) & \Gamma_{rt}^r &= \frac{-M}{r^2} \left(1 - \frac{2M}{r}\right) = -\Gamma_{tr}^t \\ \Gamma_{r\theta}^\theta &= \frac{1}{r} = \Gamma_{r\phi}^\phi & \Gamma_{\theta\theta}^r &= -r \left(1 - \frac{2M}{r}\right) \\ \Gamma_{\phi\phi}^r &= -r \sin^2 \theta \left(1 - \frac{2M}{r}\right) & \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta \\ & & \Gamma_{\theta\phi}^\phi &= \cot \theta\end{aligned}$$

- To compare Newtonian and general relativistic trajectories, far from the body producing gravitational field, coordinate length and time become proper length and time due to asymptotic flatness; in otherwords, we need to compare $\frac{d^2 r}{dt^2}$ for GR using geodesic equation where r is quite large. Using

$$\frac{d^2 r}{d\tau^2} = \frac{d^2 r}{dt^2} \dot{t}^2 + \frac{dr}{dt} \ddot{t}$$

the geodesic equations for r and t become

$$\begin{aligned}\ddot{r} + \Gamma_{rr}^r \dot{r}^2 + \Gamma_{tt}^r \dot{t}^2 &= 0 \\ \ddot{t} + \Gamma_{tr}^t \dot{r} \dot{t} &= 0\end{aligned}$$

Plugging and simplifying and reinserting G, c ,

$$\frac{d^2 r}{dt^2} = -\frac{GM}{r^2} + \frac{2G^2 M^2}{r^3 c^2}$$

so in the newtonian limit $t \rightarrow \tau$ and $c \rightarrow \infty$ we recover regular equation of motion for Newtonian gravity. Thus, the parameter M in the Schwarzschild solution is mass in the Newtonian limit.

3.5 Newtonian orbits

In the following discussion, we work in polar coordinates. By conservation of energy, for a Newtonian orbit $E_n = K + V = \frac{1}{2}mv^2 - \frac{GMm}{r}$ is conserved. The equation of motion in r , with $\dot{\theta}$ expressed in terms of conserved angular momentum $l = mr^2\dot{\theta}$, involves only r and its derivatives. It is the same equation as would be obtained for a fictitious one-dimensional problem in which a particle of mass m is subject to a force $f' = f + \frac{l^2}{mr^3}$. The significance of the additional term is clear if it is written as $mr\dot{\theta}^2 = \frac{mv_\theta^2}{r}$, which is the familiar centrifugal force. Then the equation of motion of the particle in r is that of a one-dimensional problem with a fictitious potential $V' = V + \frac{1}{2} \frac{l^2}{mr^2}$. So the effective potential energy is $V = -\frac{GMm}{r} + \frac{l^2}{2mr^2}$; plotting this against r gives a potential well. At the bottom of this potential well are circular orbits. At energies where a horizontal line intersects the plotted curve at 2 points, we have elliptic orbits. If it intersects at only one point, we have parabolic/hyperbolic orbits; unless they are directly headed towards each other, the masses pass each other, never to meet again.

3.6 Measurements

- The energy of a particle measured by the observer moving with four-velocity u^μ is $E = -p_\mu u^\mu$.
- For massless particles, since our wave four-vector k^μ is related to p^μ by some constant, we can replace $p^\mu \rightarrow k^\mu$ and $E \rightarrow \omega$, yielding $\omega_0 = -k_\mu u^\mu$.

3.7 Schwarzschild geodesics

- In the Schwarzschild metric, $L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} = \sqrt{-f(r)\dot{t}^2 + \frac{\dot{r}^2}{f(r)} + r^2\dot{\phi}^2}$ where $\dot{t} = \frac{dt}{d\lambda}$ and $f(r) = (1 - \frac{2M}{r})$.
- One can take a shortcut by using the Lagrangian $\mathcal{L} = -f(r)\dot{t}^2 + \frac{\dot{r}^2}{f(r)} + r^2\dot{\phi}^2$ where we parametrize directly using proper time i.e. $\dot{t} = \frac{dt}{d\tau}$. Take $\mathcal{L} = \epsilon$ where ϵ is a constant of motion, 0 for null geodesics and -1 for timelike geodesics, and the geodesic equation is $\frac{\partial \mathcal{L}}{\partial x^\mu} = \frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right)$.
- Generalized momenta are defined similarly to in the Lagrangian approach. For timelike geodesics, $\frac{p_\mu}{m} = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \Big|_{L=1, \tau=\lambda}$ where m is the mass of the particle.
- In general, we normalize four-velocity so $L^2 = -1$ and $L = 1$.
- Since $\frac{\partial \mathcal{L}}{\partial t} = 0$, energy per unit rest mass is conserved, that is, $\frac{p_t}{m} = -E = -f(r)\frac{dt}{d\tau} = -f(r)\dot{t} = \text{constant}$.
- Similarly, since $\frac{\partial \mathcal{L}}{\partial \phi} = 0$, momentum per unit mass is conserved. That is, $\frac{p_\phi}{m} = J = r^2\dot{\phi} = \text{constant}$.
- So the conserved quantities are $\mathcal{L}$, E , and J .
- From the constants of motion,

$$\frac{E^2 - 1}{2} = \frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 - \frac{M}{2} + \frac{J^2}{2r^2} - \frac{MJ^2}{r^3}$$

where effective potential

$$V_{eff} = -\frac{M}{r} + \frac{J^2}{2r^2} - \frac{MJ^2}{r^3} = -\left(\frac{r}{M}\right)^{-1} + \frac{1}{2} \left(\frac{J/M}{r/M}\right)^2 - \frac{(J/m)^2}{(r/M)^3}$$

To compare with Newtonian orbits let's reinstate c, G and write E as the energy per unit mass $\frac{E_n + mc^2}{mc^2}$ and J as the angular momentum per unit mass J_n/M . To take the Newtonian limit $c \rightarrow \infty$ we recover our usual Newtonian orbit expression ($\tau \rightarrow t$)

$$E_n = \frac{m}{2} \left(\frac{dr}{dt} \right)^2 + \frac{J_n^2}{2mr^2} - \frac{GMm}{r}$$

where the last two terms constitute the effective potential.

- $\frac{dV_{eff}}{dr} = 0$ when $r_{\pm} = \frac{J^2}{2M} \left[1 \pm \sqrt{1 - 12 \left(\frac{M}{J} \right)^2} \right]$; $r_+ = r_{min}$ is where stable circular orbit occurs. (Draw here!!!!)